

5.7 Lifting instructions

For lifting parts and assemblies of parts use lifting slings, working load limit (WLL): 300 kg.

Lifting the separator.

1. Remove the separator bowl.
2. Attach endless slings or cables.
3. Lift and handle with care.

NOTE

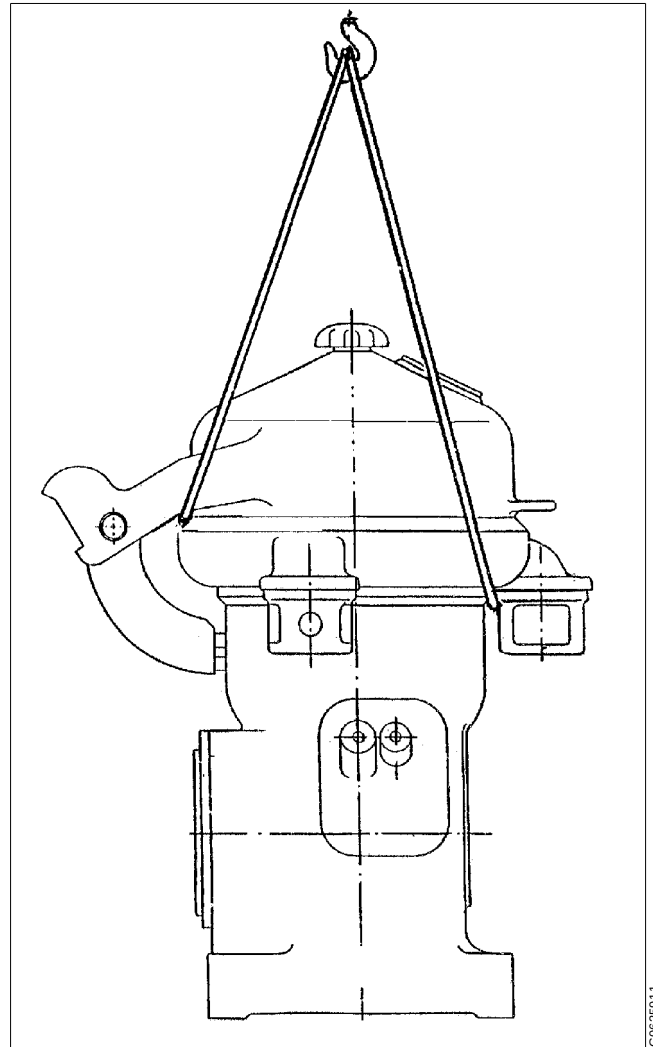
Remove the separator bowl before lifting to prevent bearings to be damaged.



WARNING

Crush hazards

A dropped separator can cause accidents resulting in serious injury to persons and damage to equipment.



Do not lift the separator in any other way

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6 *Dismantling/Assembly*

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6.1 General

The parts must be handled carefully. Don't place parts directly on the floor, but on a clean rubber mat, fibreboard or a suitable pallet.

6.1.1 References to check points

In the text you will find references to the check point instructions in Chapter 5. The references appear in the text as in the following example:

✓ **Check point**

“5.3.4 Disc stack pressure” on page 62.

In this example, look up check point “5.3.4 Disc stack pressure” on page 62 for further instructions.

6.1.2 Tools

Special tools from the tool kit shall be used for dismantling and assembly. The special tools are specified in the *Spare Parts Catalogue*.

For lifting parts and assemblies of parts use lifting slings, working load limit (WLL): 300 kg.

6.2 Dismantling

- To avoid accidental start, switch off and lock power supply.

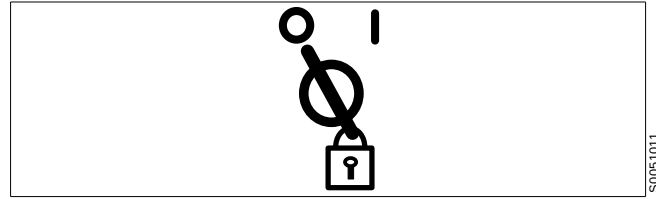


DANGER

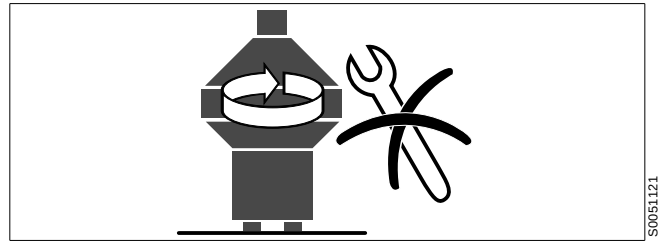
Entrapment hazards

Make sure that rotating parts have come to a **complete standstill** before starting **any** dismantling work.

The revolution counter indicates if separator parts are rotating or not.



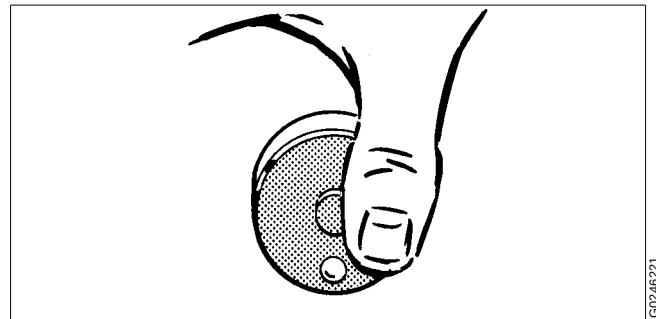
Lock power supply



Do not dismantle any parts before complete standstill

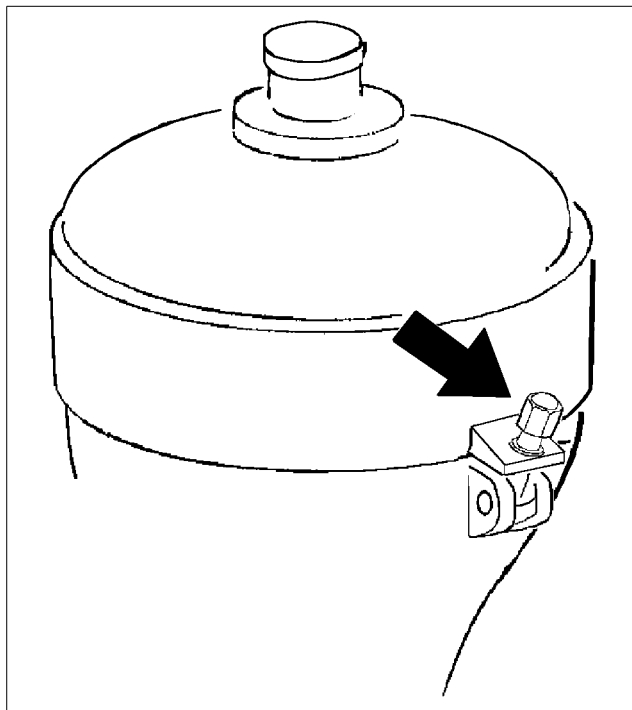
6.2.1 Bowl

1. Be sure the bowl has stopped rotating



Revolution counter indicates rotation

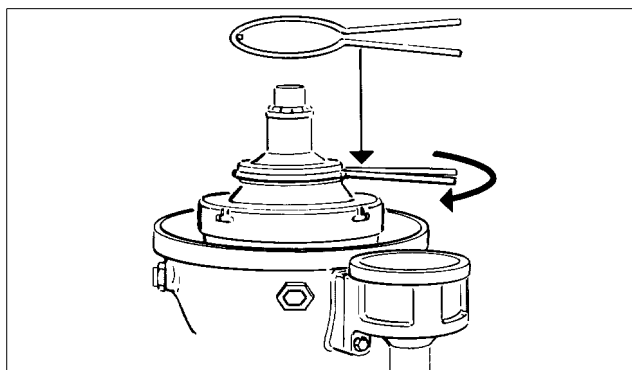
2. Loosen the hook bolts and remove the collecting cover.



Hook bolts

3. Unscrew the small lock ring by using the special tool, "Spanner for small lock ring".

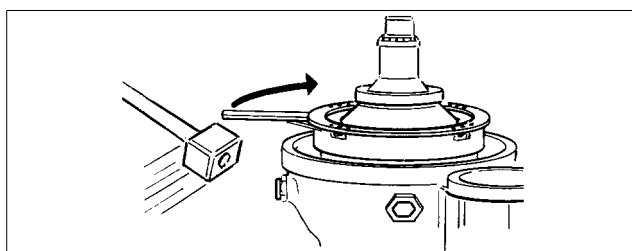
Left hand thread!



Small lock ring

4. Unscrew the large lock ring by using the special tool; "Spanner for lock ring" and a tin hammer.

Left hand thread!



Large lock ring

5. Lift off gravity disc.

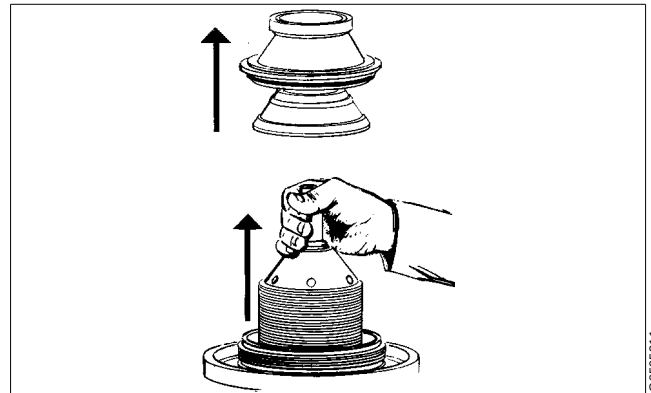
Note! If the gravity disc has to be replaced owing to changed operating conditions.
Lift out the top disc, disc stack and distributor.



CAUTION

Cut hazard

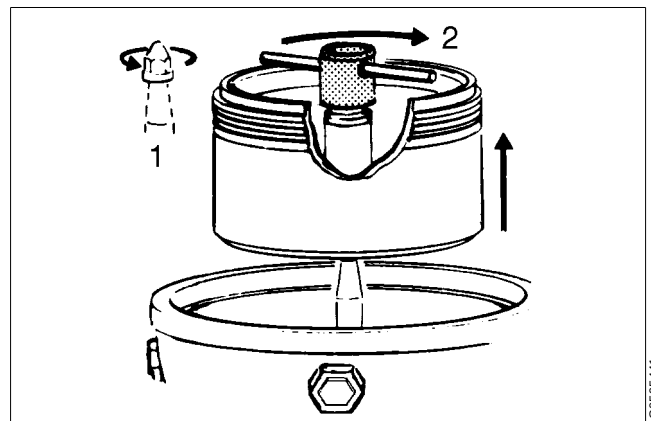
The discs have sharp edges.



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6. Unscrew the cap nut.

Lift out the bowl body using the hand tool.



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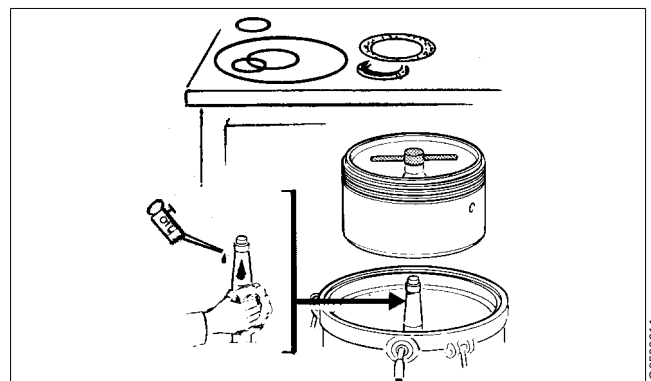
1. Cap nut
2. Lift bowl body from spindle

7. Soak and clean all parts thoroughly in suitable cleaning agent, see "5.4 Cleaning" on page 72.

Remove O-rings and replace them with spares from the Major Service kit.

✓ Check point

"5.3.1 Corrosion" on page 58,
"5.3.2 Erosion" on page 60,
"5.3.3 Cracks" on page 61.



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6.2.2 Vertical driving device

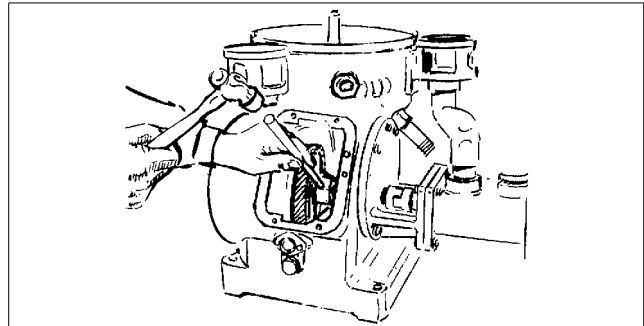
Remove the outlet housings, feed and discharge pumps and raise the collecting cover. Remove also the separator bowl.

Before dismantling, in the case of 8000 hours service, or if the separator vibrates while running:

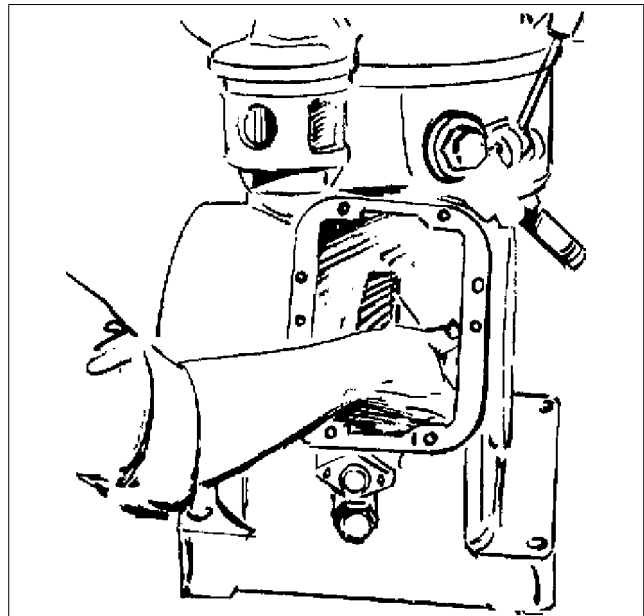
✓ **Check point**

“5.3.6 Radial wobble of bowl spindle” on page 64.

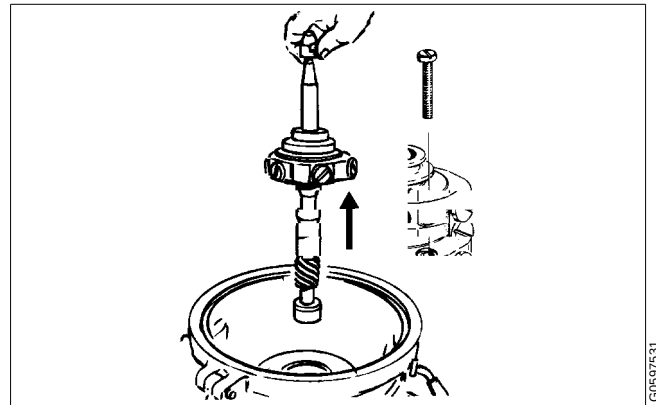
1. Drain off the oil
Force out the conical pin of the worm wheel.



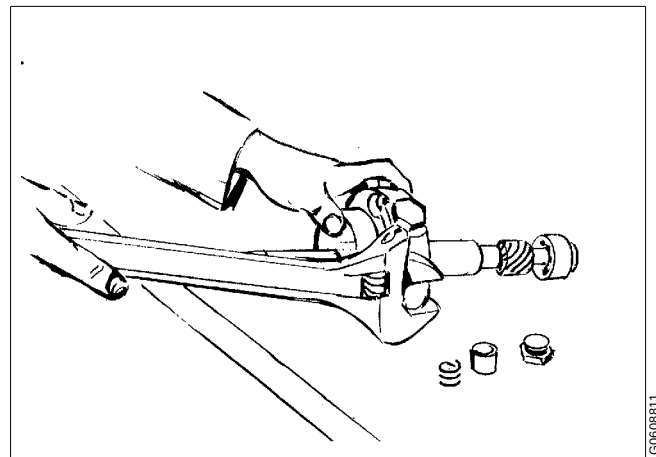
2. Push worm wheel aside.



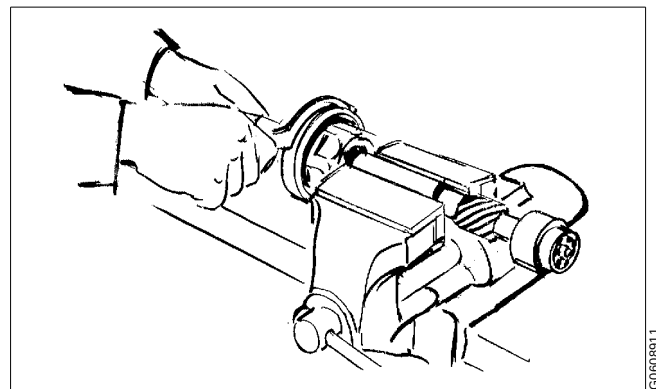
3. Loosen the spring casing.
Fit the cap nut to shaft top and lift out the vertical driving device.



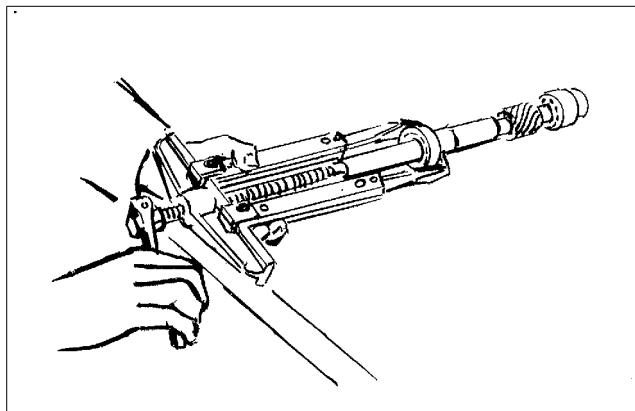
4. Loosen the buffers and remove housing.



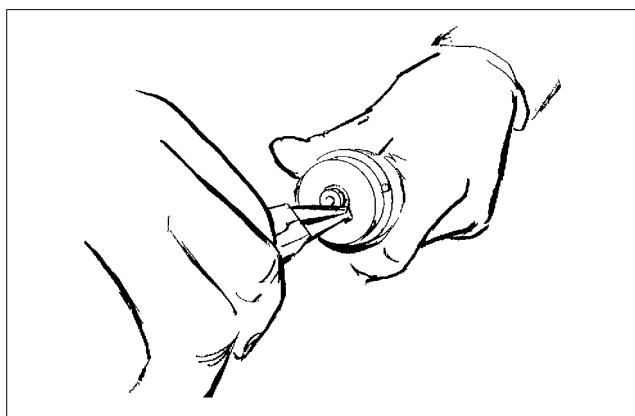
5. Remove the collar.



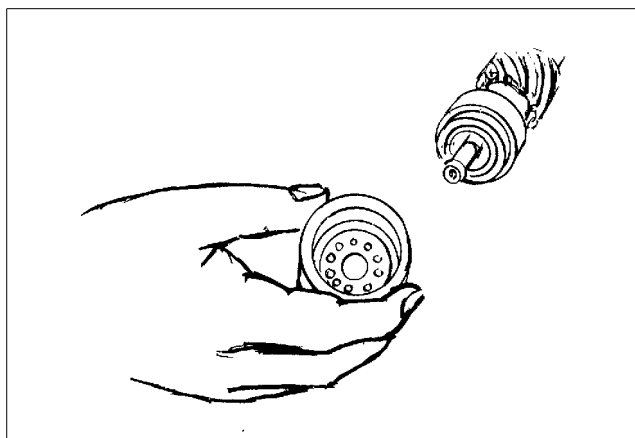
6. Use the puller tool to dismantle the upper ball bearing.



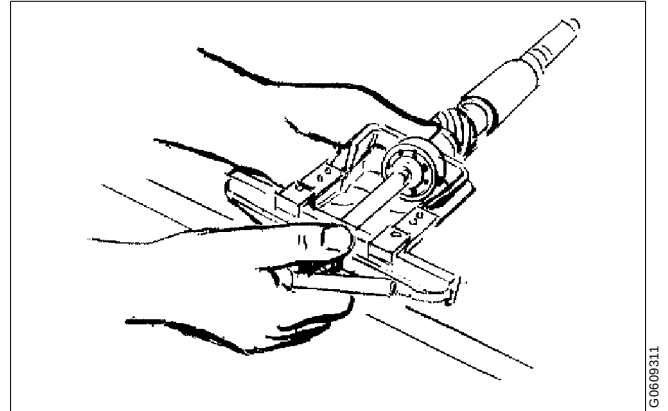
7. Remove the snap ring.



8. Remove the ball bearing housing together with thrust ball bearing and height adjusting ring(s).

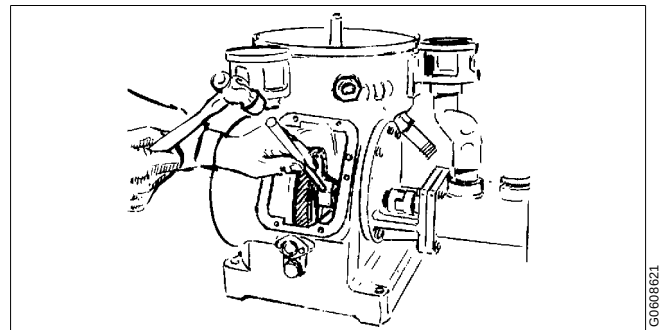


9. Pull off lower ball bearing gently.

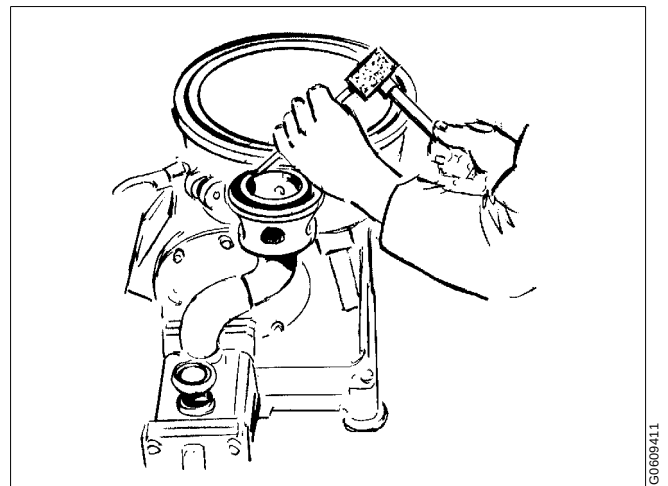


6.2.3 Horizontal driving device

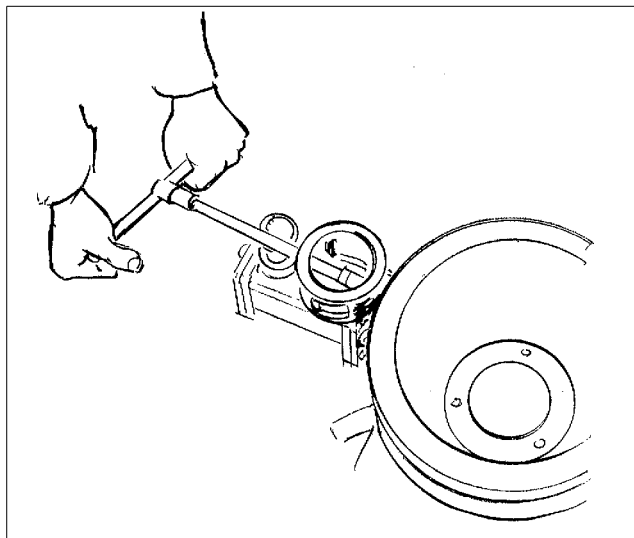
1. Drain gear housing from oil see “5.5.1 Oil change procedure” on page 76. Remove guard for worm wheel and force out the conical pin in worm wheel.



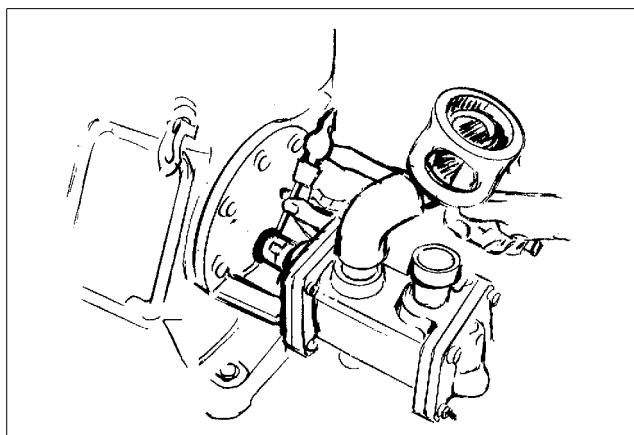
2. Remove lock ring, sight glass and seal ring (valid for machine with feed and discharge pump).
For machine without pump go to point 5.



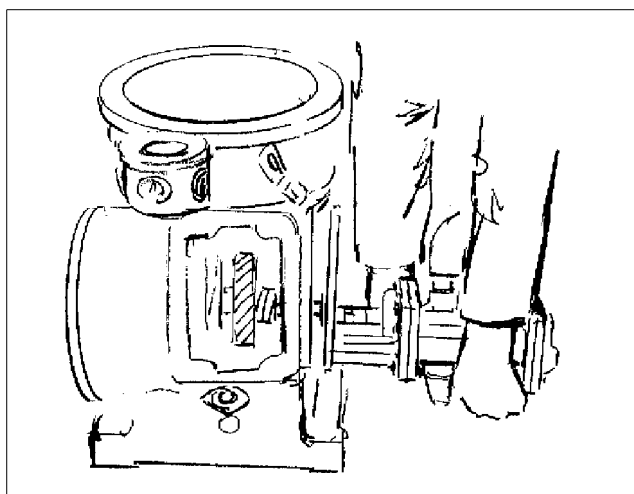
3. Loosen screws of inspection box.



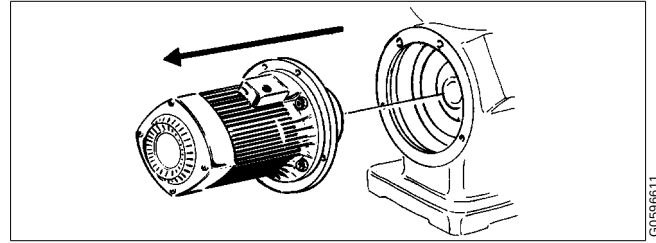
4. Remove coupling guard and force out the tubular pin in coupling half on horizontal driving shaft.



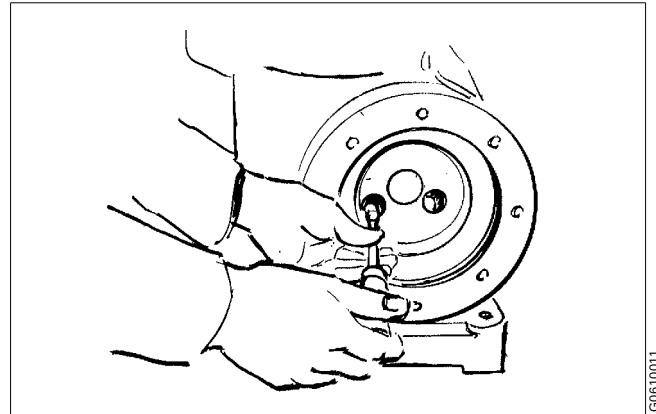
5. Loosen end shield. Remove pump together with inspection box.
For machine without pump: Loosen and remove guard. Force out the pin at shaft end and remove space sleeve.



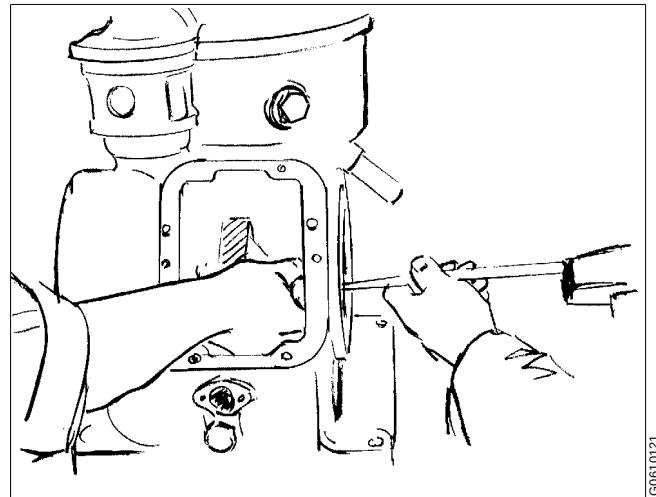
6. Remove the motor.



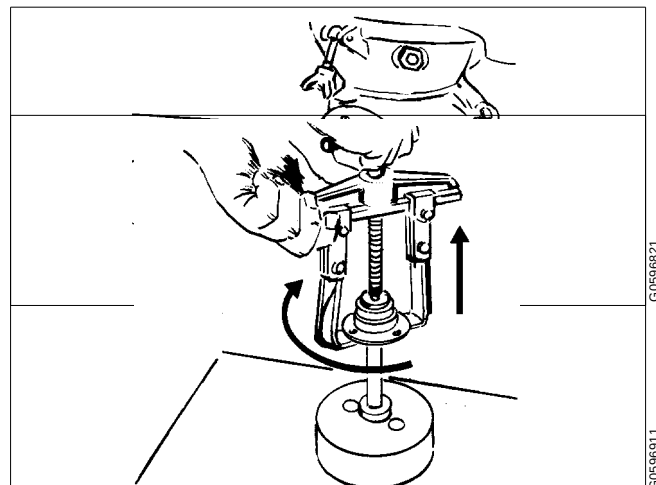
7. Loosen the three screws holding the sealing washer.



8. knock gently on shaft end to get the bearings out of their seats. Then use the worm wheel to knock off bearing from shaft. Remove bearing and worm wheel.



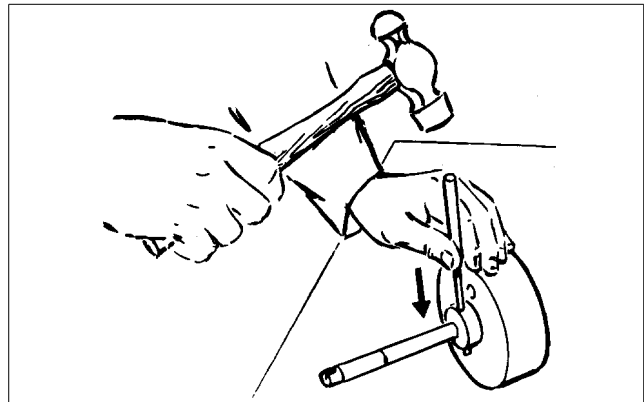
9. Remove the horizontal driving shaft



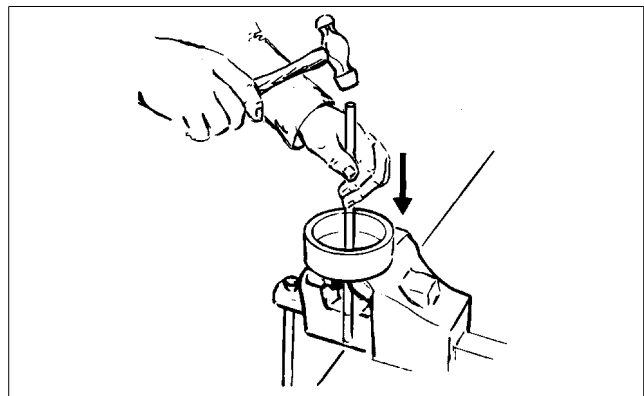
10. Use the puller tool to pull off the sealing washer.

Protect end of worm wheel shaft with a washer.

11. To replace coupling drum or worm wheel shaft force out the spring pin and gently knock out the shaft.



12. Knock out the horizontal spindle.



6.3 Assembly

Clean all parts in a degreasing agent and replace parts supplied in the Spare parts kits.

✓ Check point

“5.3.1 Corrosion” on page 58,
 “5.3.2 Erosion” on page 60,
 “5.3.3 Cracks” on page 61,
 “5.3.10 Top bearing springs” on page 68,
 “5.3.11 Ball bearing housing” on page 68,
 “5.3.12 Worm wheel and worm; wear of teeth”
 on page 69.

6.3.1 Vertical driving device

Before assembling the bowl spindle, make sure the relevant checks have been carried out.

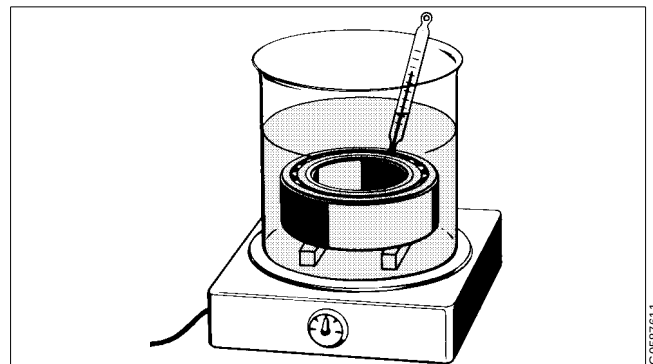
Wipe off and oil the bearing seat before fitting the ball bearing.

Inspect the tapered end of the bowl spindle and assemble ball bearings.

Heat the new ball bearings in oil to maximum 100 °C.

NOTE

Always fit new bearings.



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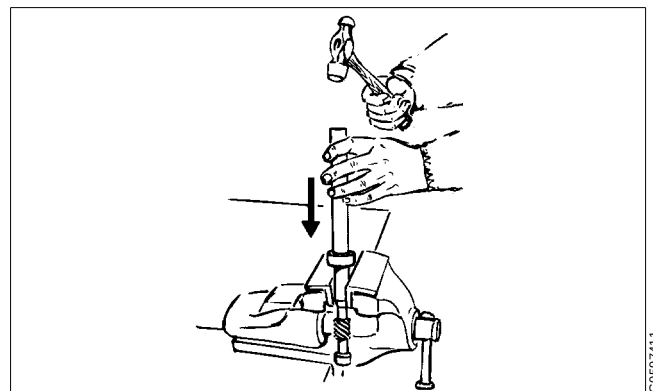
1. Fit the two bearings onto the shaft.



CAUTION

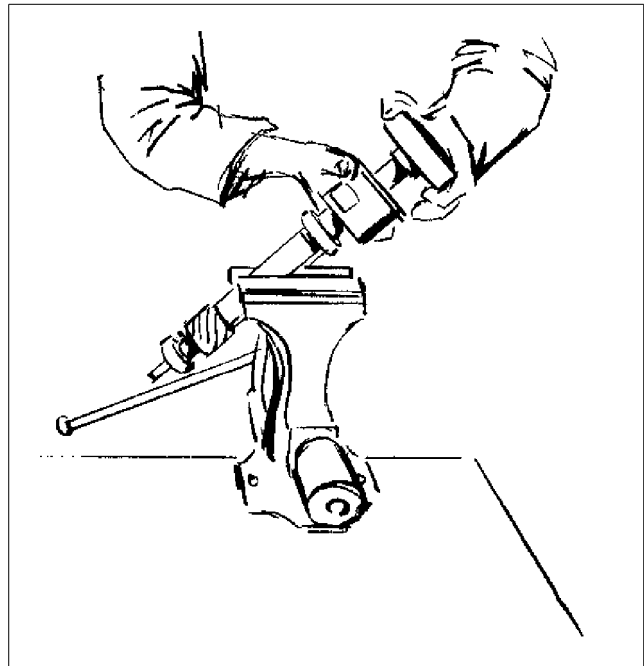
Burn hazards

Use protective gloves when handling the heated bearings.

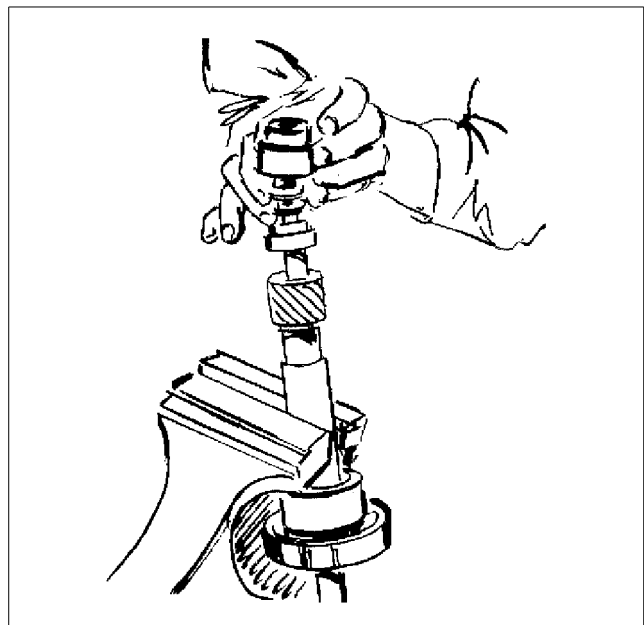


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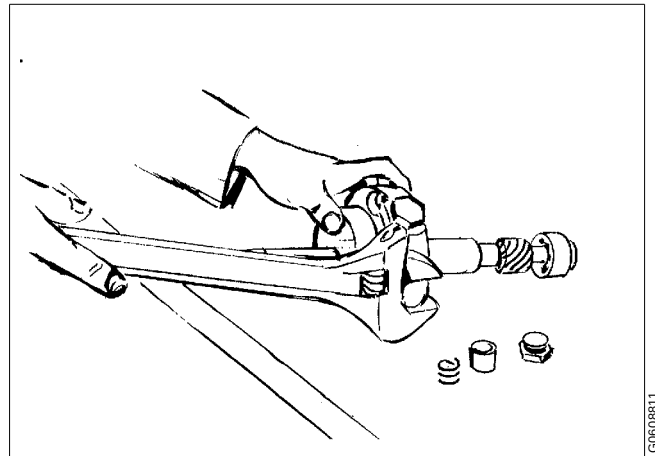
2. Fit ball bearing housing and throw-off collar (left hand thread).



3. Fit the ball bearing housing together with thrust bearing and height adjusting ring(s). Secure with snap ring.



4. Place spring casing to rest on shoulder of upper ball bearing housing. Fit buffers and springs into casing, Tighten the caps.

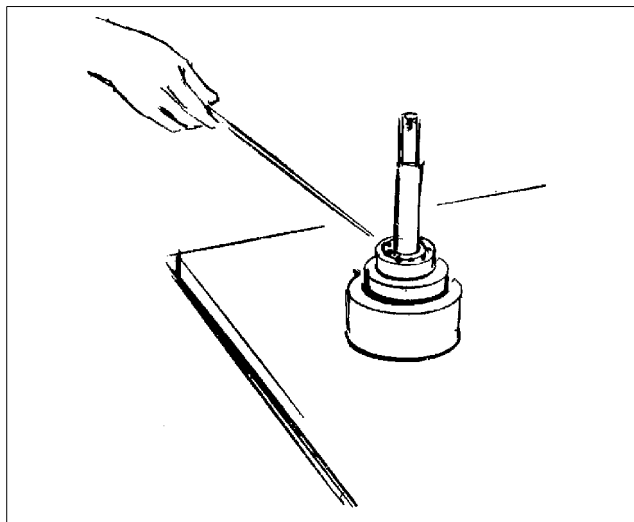


5. Remember to fit the packing



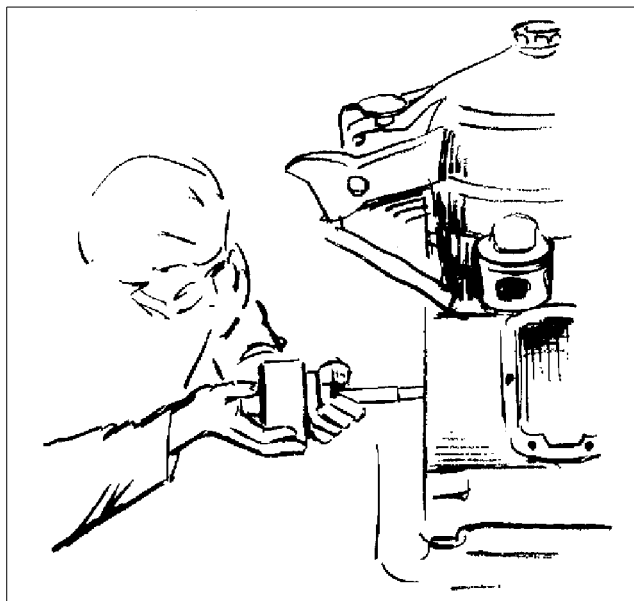
6.3.2 Horizontal driving device

1. Fit protection cover on worm wheel shaft.
Heat bearing in oil before mounting onto shaft.



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2. Clean the ball bearing housing in the frame and oil the outer race of the ball bearing.
Just enter the shaft through the bearing seat in frame.



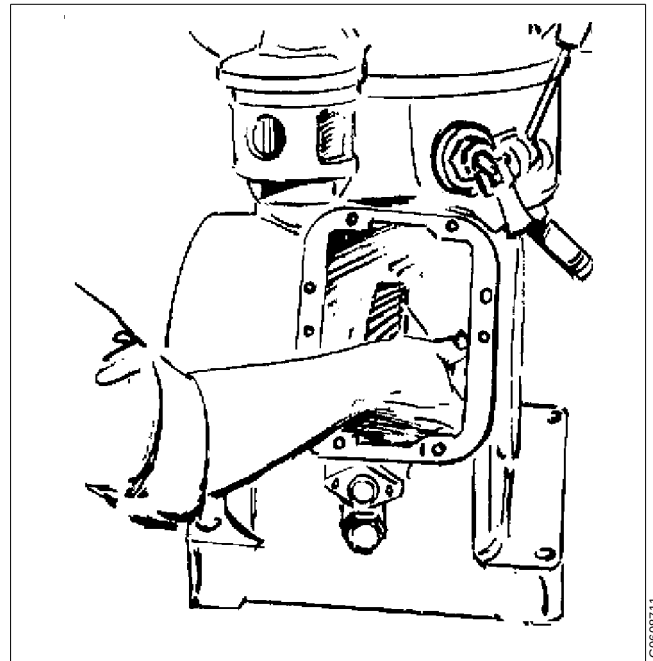
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3. Enter the worm wheel on shaft and force the worm wheel shaft to its position in the frame, so that the ball bearing enters correctly into its seat.

Observe that largest ends of the holes for conical pin in shaft and worm wheel should point in same direction.

**DANGER****Disintegration hazard**

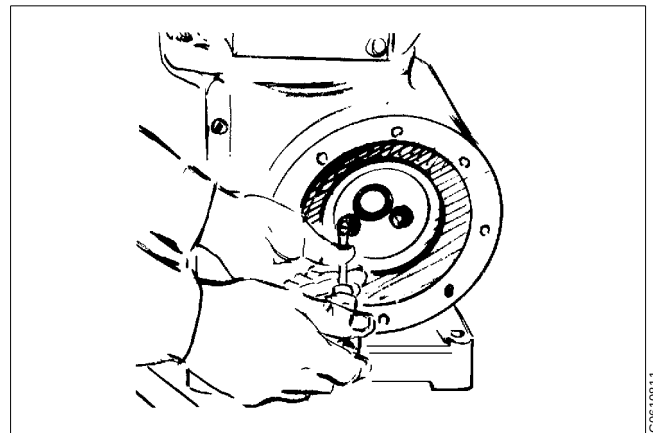
When replacing the gear, always make sure that the new worm wheel and worm have the same number of teeth as the old ones.



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4. Secure the protection cover with the three screws.

Fit the 'loose' ball bearing. Secure the worm wheel with the conical pin.



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5. Lift the motor in position and tighten the motor bolts.

If the coupling disc has been loosened without first marking its position on the motor shaft, the correct position for position of coupling disc must be determined again.

✓ **Check point**

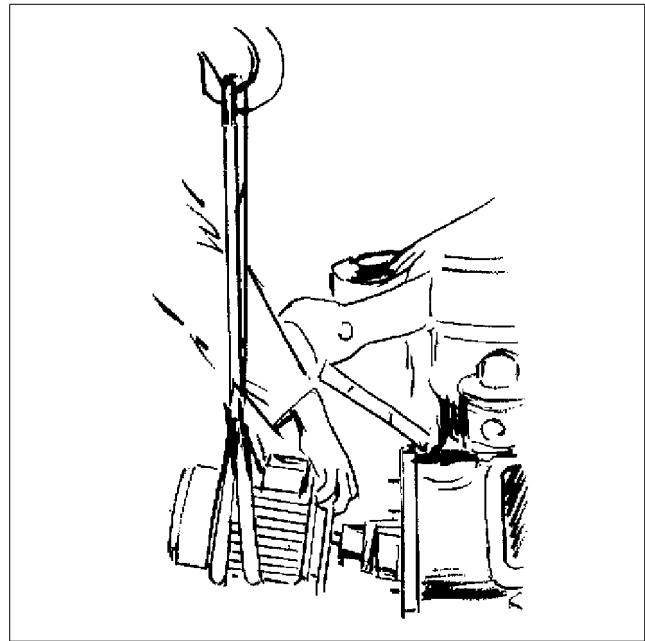
"5.3.9 Coupling disc of motor" on page 67.



DANGER

Disintegration hazards

When power cables have been connected, always check direction of rotation. If incorrect, vital rotating parts could unscrew causing disintegration of the machine.

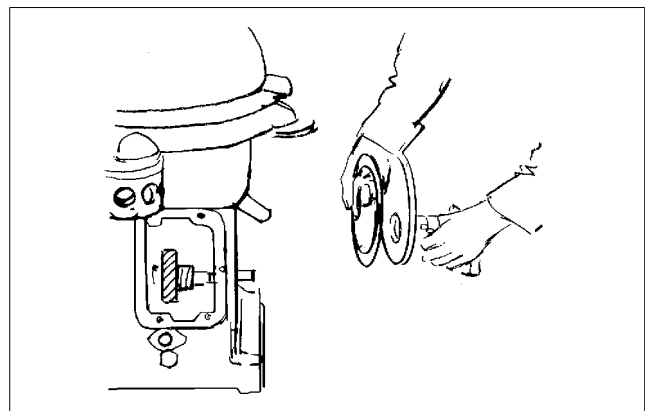


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6. Fit coupling halve on worm wheel shaft before mounting the pump. Remember the gasket.

Fit the coupling guard with two screws.

For machine **without pump** fit distance sleeve and pin at the end of horizontal shaft. Fit and secure the guard.



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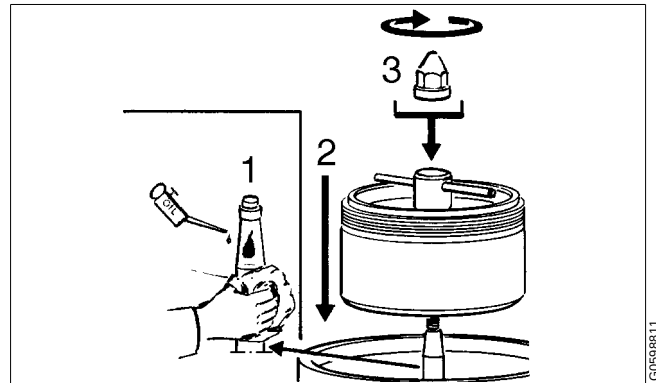
6.3.3 Bowl

Make sure that the check points are carried out before and during assembly of the separator bowl.

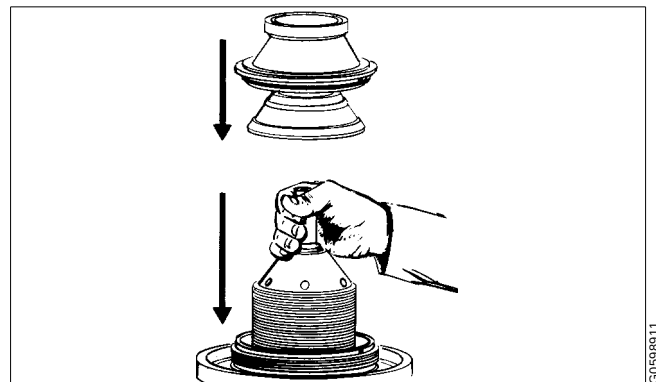
✓ **Check points**

“5.3.8 Bowl spindle cone and bowl body nave” on page 66.

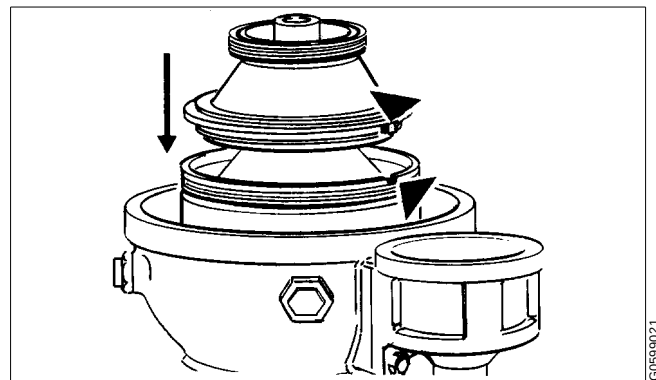
1. Wipe off spindle top and nave bore in the bowl body. Apply a few drops of oil onto the taper, smear it over the surface and wipe it off with a clean cloth.



2. Fit the distributor with the disc stack.
Fit the top disc and bowl hood.

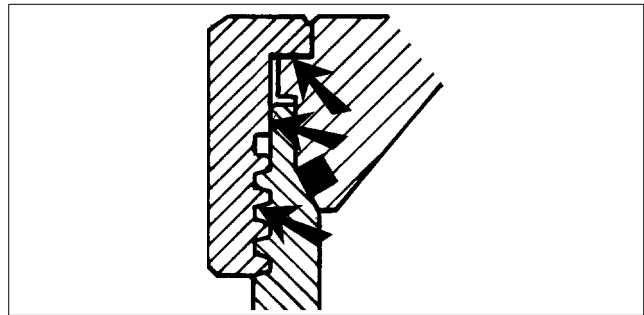


3. Make sure the guides are in correct position.

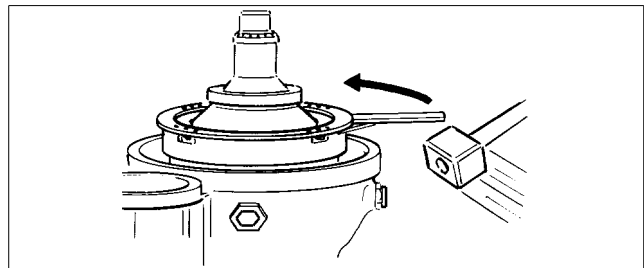


4. Degrease lock ring threads, contact and locating faces (see arrows above). Apply Molykote 1000 paste to the threads and faces stated.

Brush in the paste according to the manufacturer's direction.



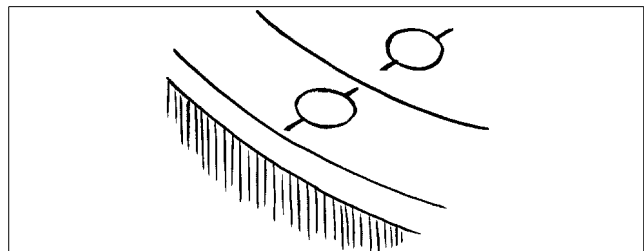
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5. Tighten lock ring counter-clockwise (left-hand thread) until bowl hood lies tightly against bowl body (in a new bowl marks will now be in line with each other).

Left-hand thread!



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WARNING

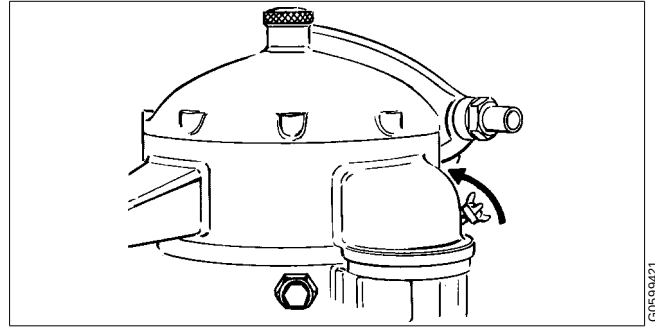
Disintegration hazard

The assembly mark on the lock ring must never pass the mark on the bowl hood by more than 25° (or 45 mm).

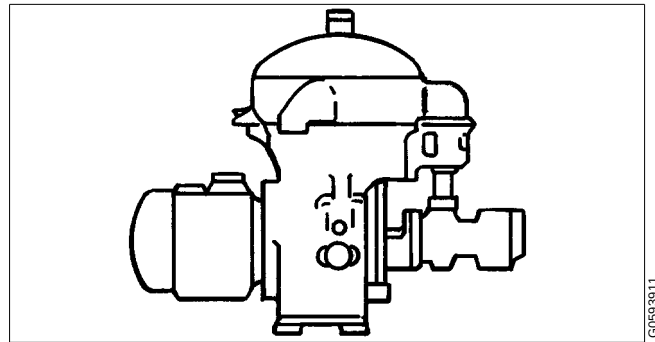
✓ Check point

"5.3.4 Disc stack pressure" on page 62.

6. Tighten clamp bolts.



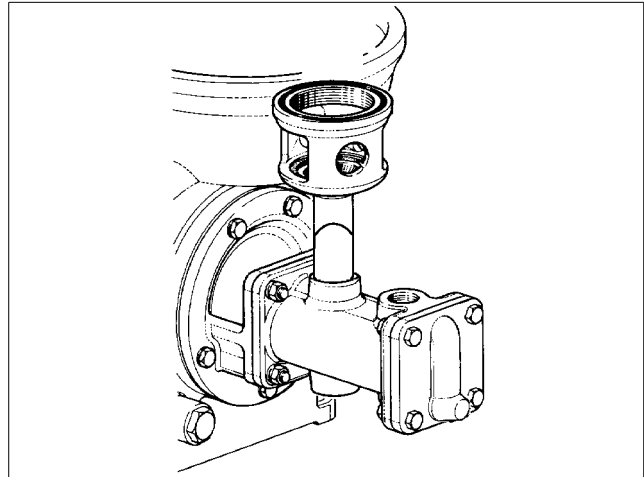
7. Fit the connecting hoses if they have been removed. Make sure to fit their gasket rings



6.4 Feed and discharge pumps

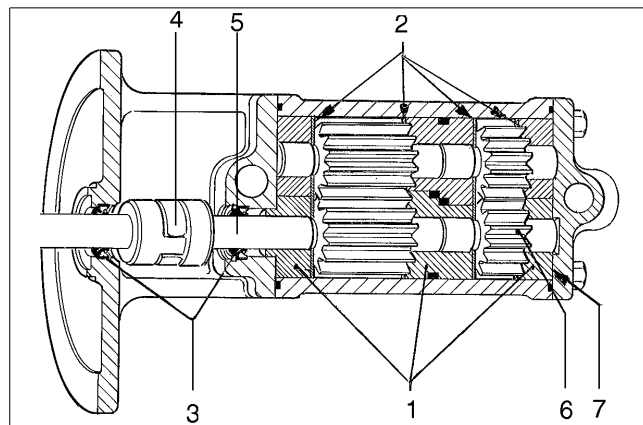
6.4.1 General description

1. Relief/safety valve:
Examine valve cone and valve seat.
2. Bushings:
Exchange the bushings if they are scratched or there is a play between shaft and bushing.
3. Wearing seals:
Replace the seals if the surface is rough, crackled or dented by the impeller.
4. Lipseal rings:
Replace the rings at the annual overhaul.
Important! Turn the rings the right way round.
5. Flexible coupling:
Check wear on flexible element.
6. Impeller shaft:
Check wear on sealing surface on the impeller shaft.
7. Disengagement:
The feed pump can be disengaged by turning the impeller (6), thereby placing the driving blade of the impeller in the recess of the shield.



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Double pump



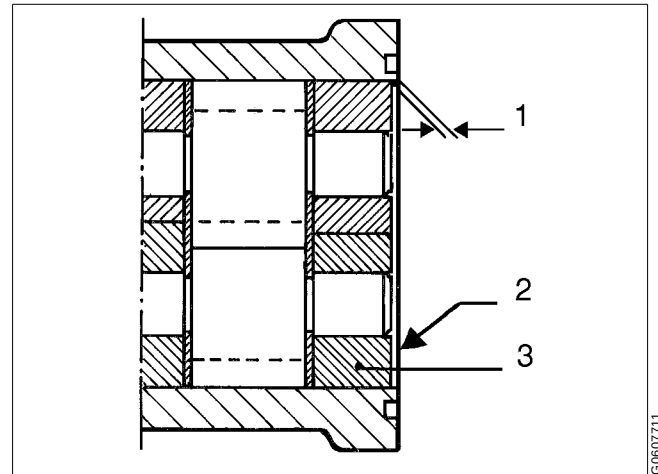
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1. Bushings
2. Wearing seals
3. Lipseal rings
4. Flexible coupling
5. Impeller shaft
6. Disengagement
7. Axial play

8. Axial play:

The total axial play (1) must be 0,1 - 0,3 mm. If the play is too large even though the wearing seals have been renewed, it can be compensated by adding a brass leaf liner.

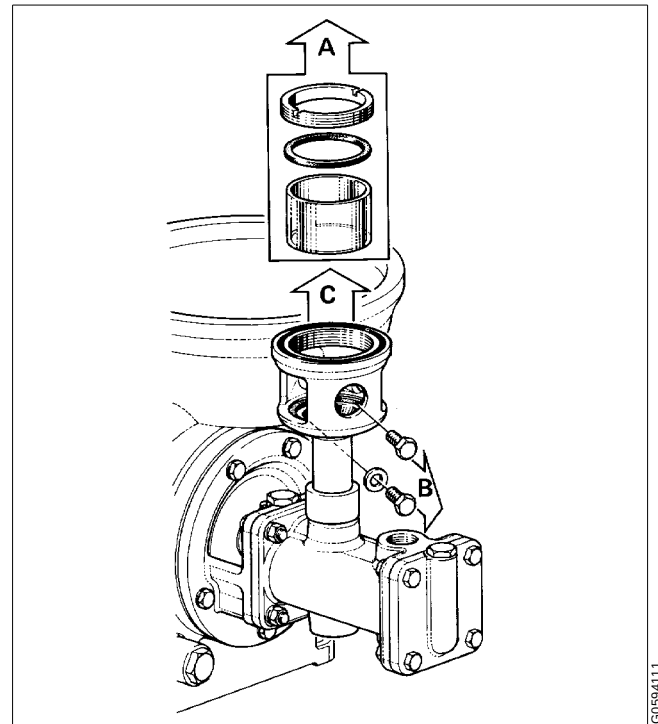
Insert the liner at (2). If the play is too small, grind off the bearing holder (3).



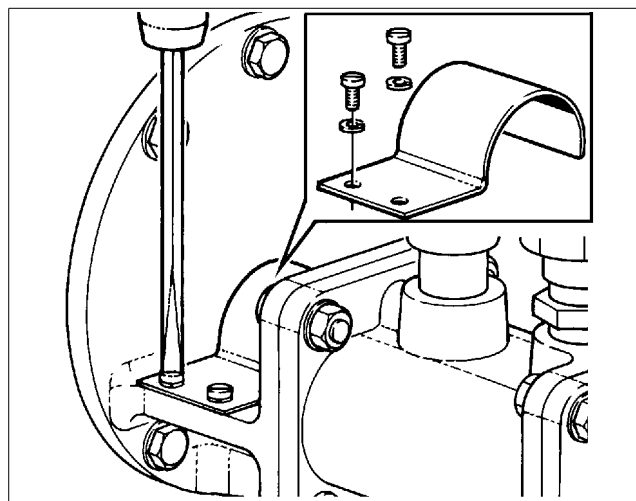
Axial play 0,1 - 0,3 mm

6.4.2 Dismantling

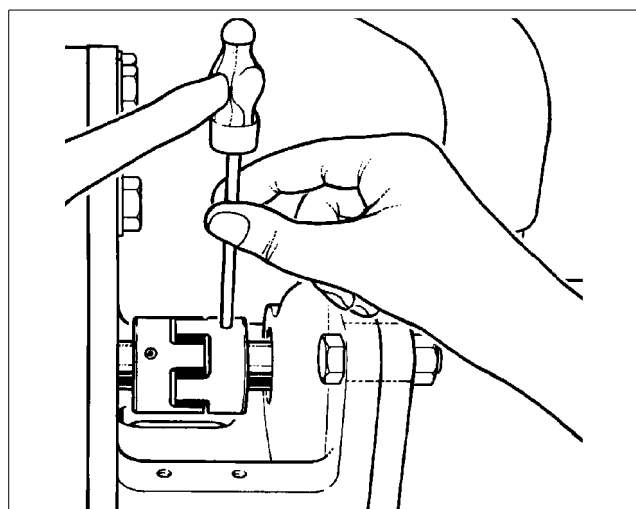
1. Remove the pipe connections of the pump.
Screw off the lock ring of the sight glass.
Remove the upper gasket and the sight glass.
Remove the screws of the control housing.
Lift the control housing with the lower gasket and the connecting piece.
Unscrew the oil drain plug and empty the gear housing from oil.



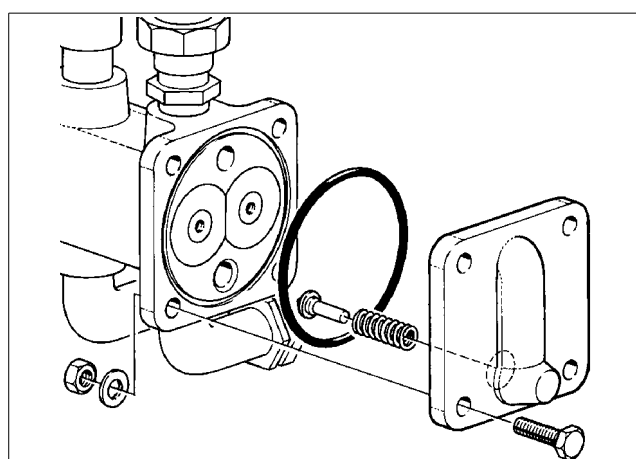
2. Remove the guard over the flexible coupling.



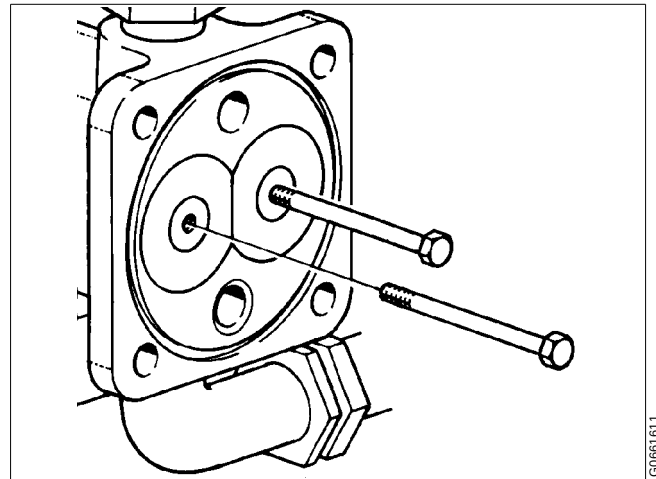
3. Drive out the tubular pins from the worm wheel shaft and impeller shaft. Turn the bowl by hand until the pins are in a convenient position.



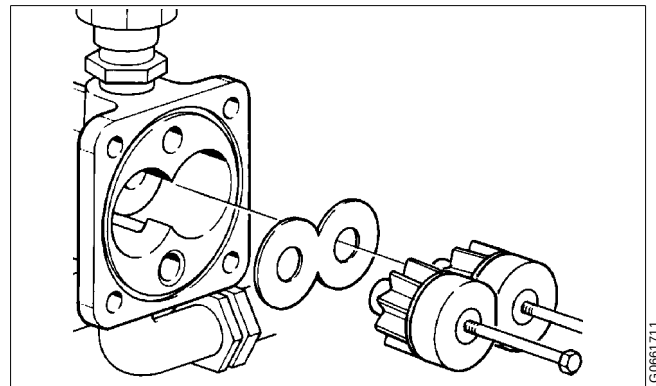
4. Undo the four screws holding the pump shield, remove shield.
Do not lose the valve cone and spring.



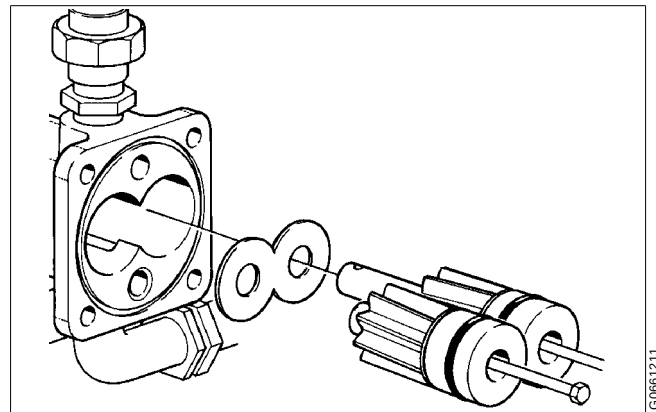
5. Insert two screws into the center hole of the impellers. Use the screws as a handle to withdraw the impellers.



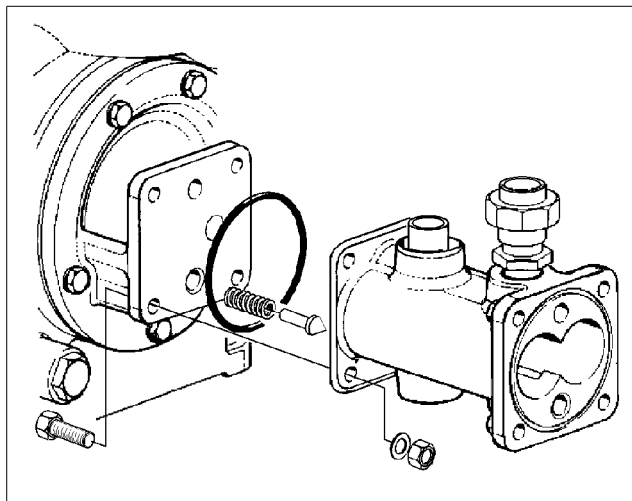
6. Remove the impellers for inlet pump.



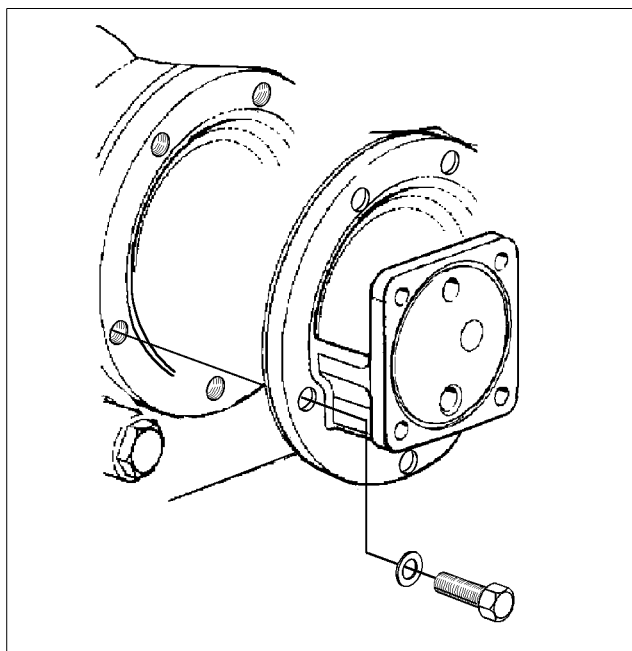
7. Repeat step 5 and 6 for the outlet pump impellers.



8. Undo the four screws holding the pump housing. Remove housing, do not lose valve cone and spring.

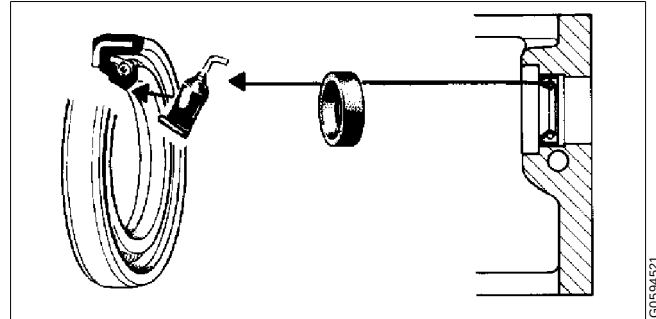


9. Check that the lipseal rings are faultless. Should this not be the case, remove the pump shield from separator frame and knock out the faulty sealring(s).

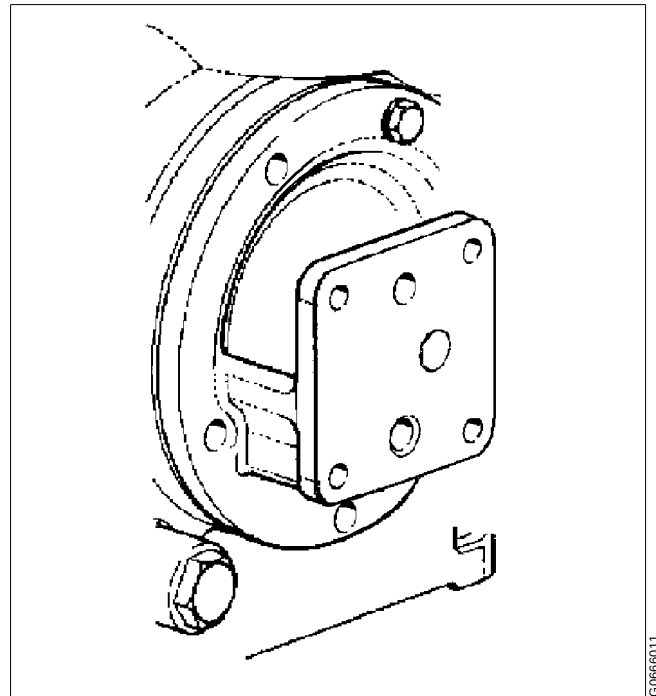


6.4.3 Assembly pump

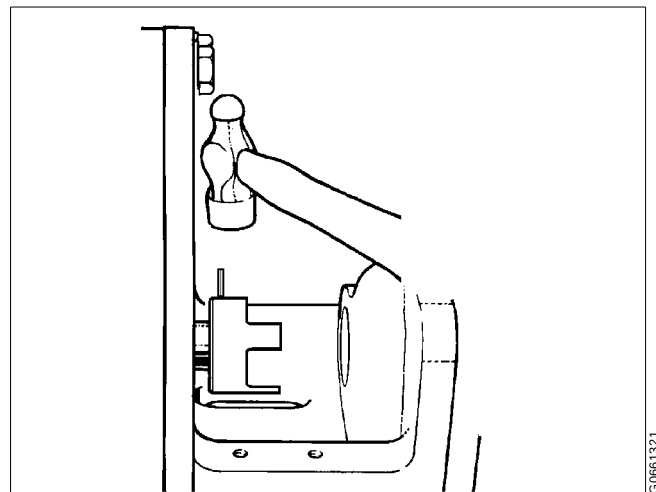
1. Fit new lipseal rings. Be sure to turn it the right way around, the closed side should point outwards to air. Start with the one at pumpside of the shield.



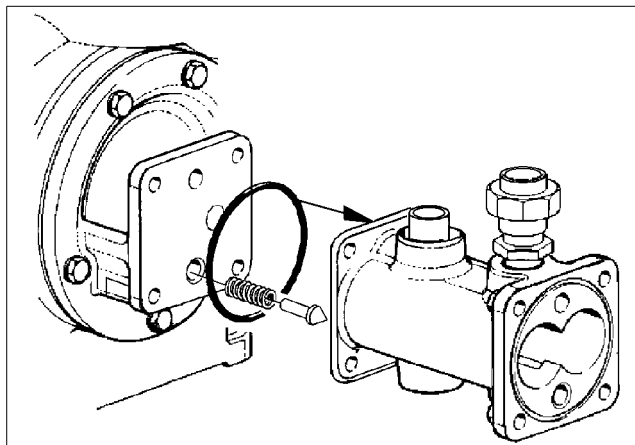
2. Lift pumpshield in place and secure it with one screw at top of shield, do not tighten it fully. Be careful not to damage the seal ring when entering the worm wheel shaft.



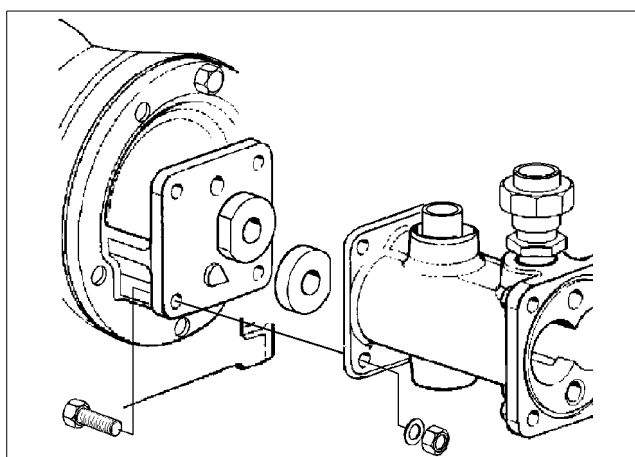
3. Fit and secure the coupling half with the larger pinhole on worm wheel shaft. Put the flexible element in place.



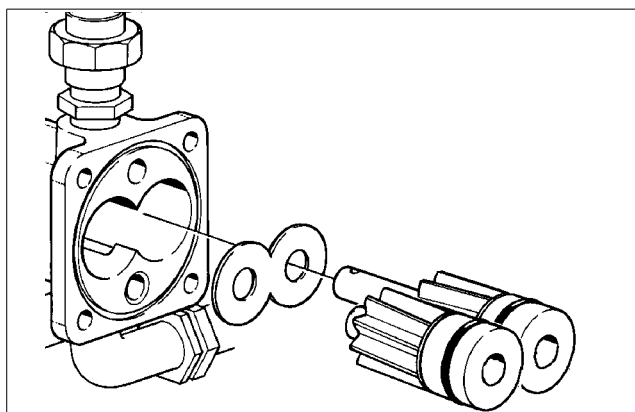
4. Place the O-ring in pump housing.
Put the spring and valve cone in place in the pump shield.



5. Place the bushing with shoulder in pump shield (flat side vertical pointing to center).
Enter the other bushing into pump housing with flat side to center.
Fit housing by entering over the bushing in shield. Ensure that valve cone enters the valve seat.
Fit and tighten the screws.

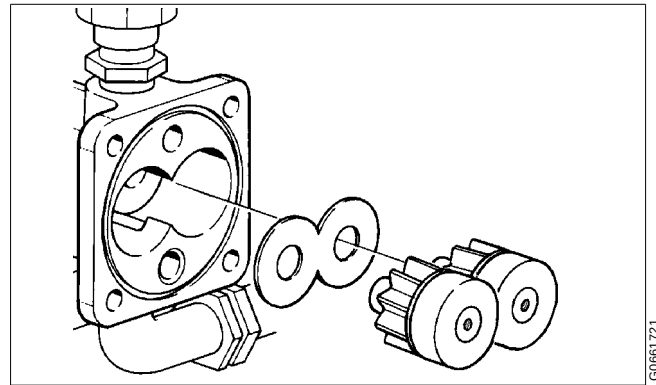


6. Put the wear gaskets one at each side of the outlet impellers, then place the bushings one in opposite direction to the other on impeller shaft.
Keep all parts together and push the package in to pump housing.



7. Repeat point 6 for inlet impellers.
Note that the tongue on impeller shaft should fit in the slot of outlet impeller shaft.

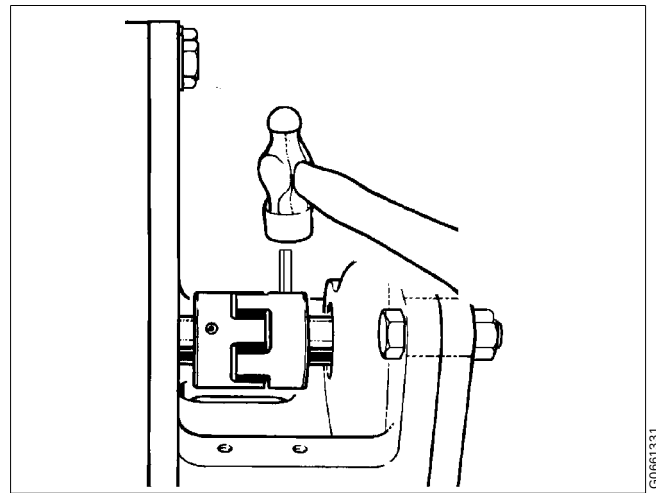
Press gently until the impeller shaft is visible at the coupling side. Enter the coupling half on shaft and press the package in place.



G0061721

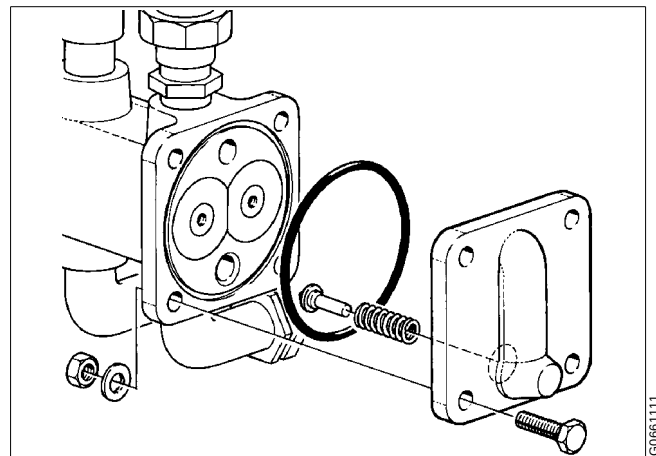
8. Fit and tighten the six screws for bearing shield.

9. Secure the coupling half with the pin.
Rotate the bowl to align the holes.



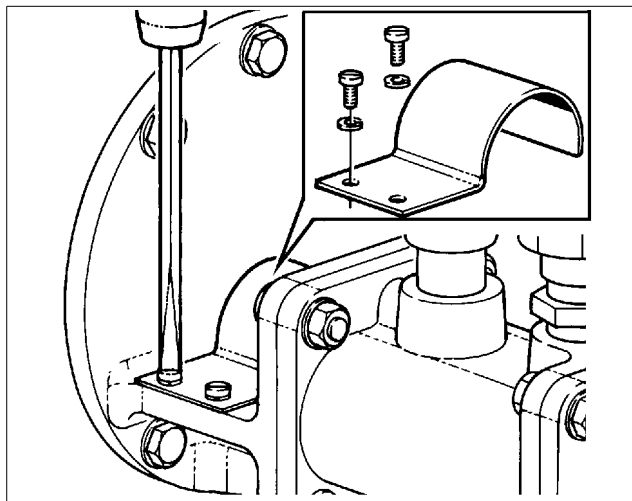
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10. Place O-ring in the groove in pump housing and the spring and valve cone in pump shield.
Fit pump shield and fasten it with four screws.

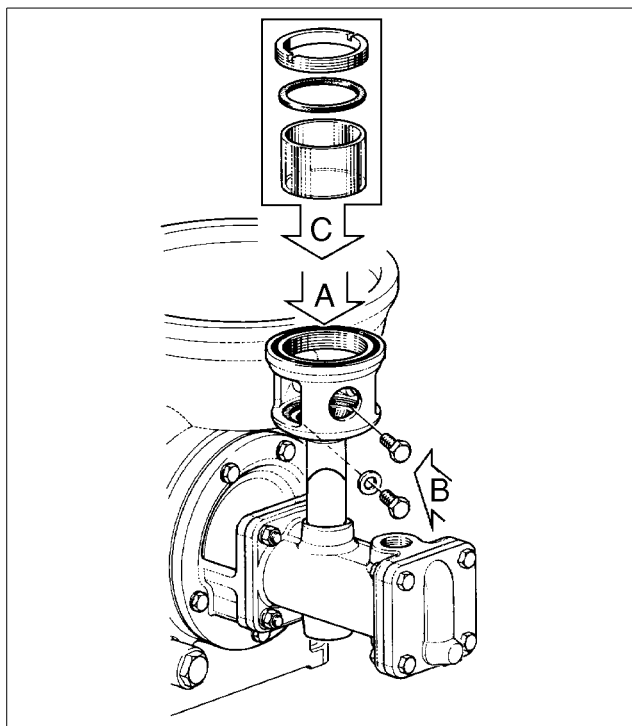


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11. Fasten the guard over the flexible coupling.



12. Fit control housing and its pipes



13. Connect all pipings.

6.5 Oil filling

NOTE

Before adding or renewing lubricating oil in the worm gear housing, the information concerning different oil groups, handling of oils, oil change intervals etc. given in chapter "8 Technical Reference" on page 125 must be well known.

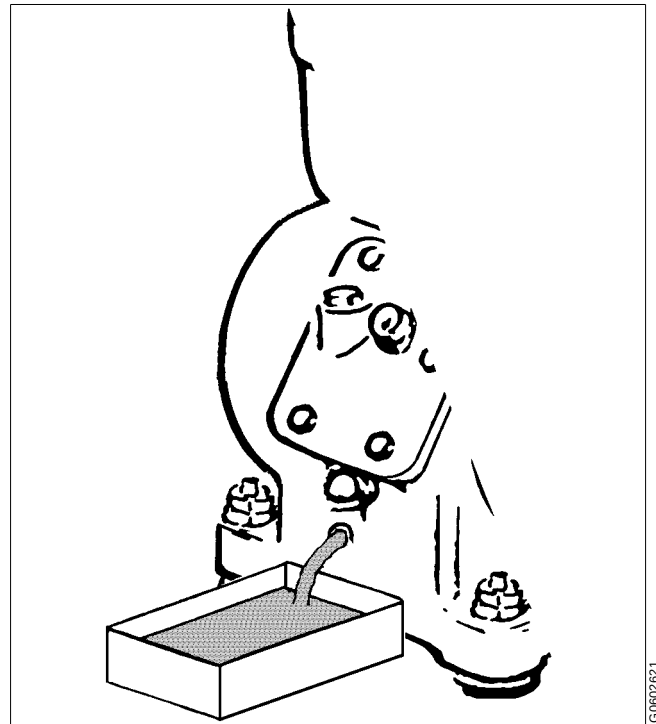
1. Place a collecting tray under the drain hole, remove the drain plug to drain off the oil.



CAUTION

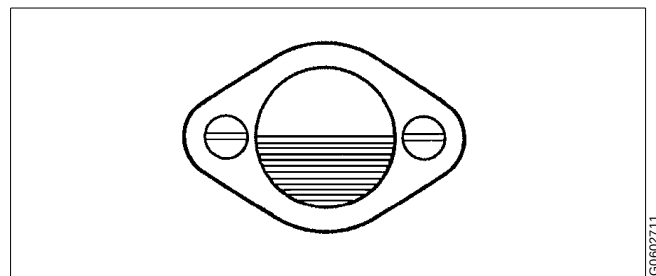
Burn hazards

Lubricating oil and various machine surfaces can be sufficiently hot to cause burns.



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2. Fill new oil in the worm gear housing. The oil level should be slightly above middle of the sight glass. For lubricating oil volume see chapter "8.1 Technical data" on page 126.



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6.6 Frame feet

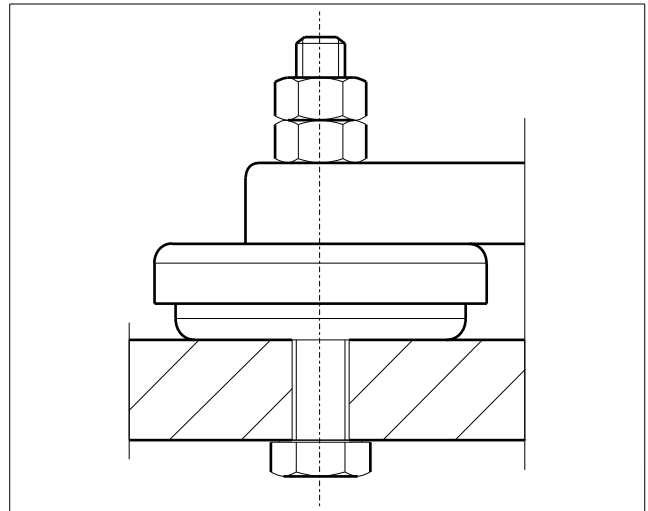
When replacing the frame feet, the separator must be lifted.

See "5.7 Lifting instructions" on page 81.

NOTE

Always remove the bowl before lifting the separator.

When lifting and moving the separator, follow normal safety precautions for lifting large heavy objects.



Frame foot with vibration damping

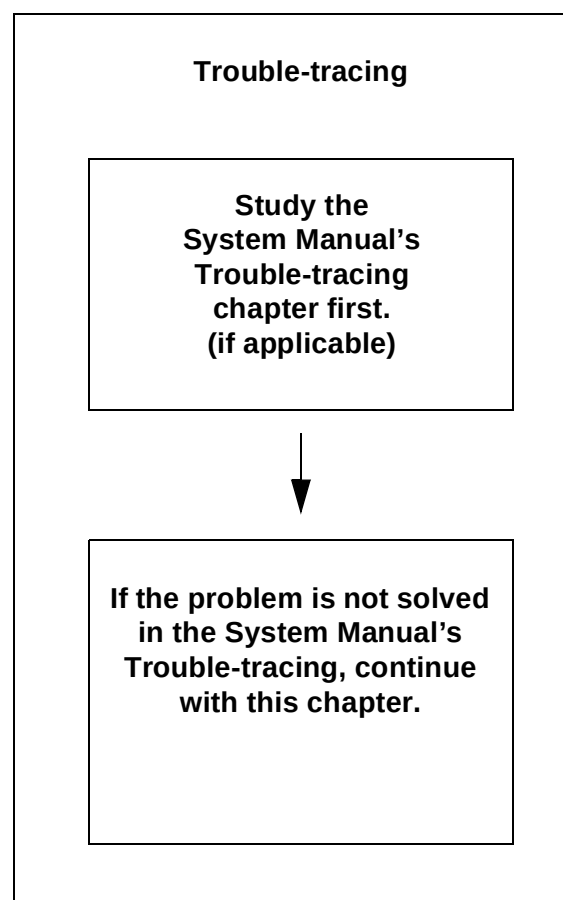
6.6.1 Mounting of new frame feet

1. Remove the existing frame feet.
2. Mount the new feet. Tightening torque 16 Nm. Secure the bolt with the lock-nut.
3. Place the separator in its original position and assemble the separator.
Remove the two eye bolts used for lifting.

7 *Trouble-tracing*

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7.1 Trouble tracing procedure

This chapter applies to trouble-tracing concerning functions of the separator only. It does not include the other equipment in your processing system.

Always start with trouble-tracing instructions in the System Manual, and if required, continue with the instructions below. If the problem still is not solved, contact your Alfa Laval representative.

7.2 Mechanical function

7.2.1 The separator does not start

Possible cause	Action
No power supply to motor.	Check power supply.

7.2.2 Start-up time too long

Possible cause	Action
Motor failure.	Repair.
Incorrect power supply (50 Hz instead of 60 Hz).	Use correct power supply. See machine plate.
Bearings damaged or worn.	Install new bearings.

7.2.3 Starting power too low

Possible cause	Action
Motor failure.	Repair the motor.

7.2.4 Starting power too high

Possible cause	Action
Motor failure.	Repair the motor.
Gear worn out.	Replace worm wheel and worm.
Bearing damaged or worn.	Install new bearings.
Incorrect transmission (50 Hz gear and 60 Hz power supply).	 DANGER: Disintegration hazard STOP immediately! Install correct transmission. Contact your local Alfa Laval representative. The bowl must be inspected.
Wrong direction of rotation.	STOP. Adjust motor power connection.

7.2.5 Separator vibrates excessively during starting sequence

NOTE Some vibration is normal during starting sequence when the separator passes through its critical speeds.

Possible cause	Action
Bowl out of balance due to: poor cleaning incorrect assembly too few discs insufficiently tightened bowl hood bowl assembled with parts from other separators.	⚠ DANGER: Disintegration hazard STOP immediately! Identify and rectify cause.
Vibration dampers in frame feet worn out.	Fit new vibration dampers.
Bowl spindle bent (max. 0,15 mm).	Fit a new bowl spindle.
Top and/or bottom bearing damaged or worn.	Fit new bearings.
Top bearing springs defective.	Fit new springs.

7.2.6 Separator vibrates excessively during normal running

Possible cause	Action
Uneven sludge deposits in sludge space.	STOP and clean bowl.
Bearings damaged or worn.	Fit new bearings.
Vibration-damping rubber washers worn out.	Fit new frame feet washers every four years.
Spindle top bearing spring(s) broken.	Replace all springs.

7.2.7 Smell

Possible cause	Action
Top and/or bottom bearing overheated.	Fit new bearings.

7.2.8 Noise

Possible cause	Action
Oil level in oil sump is too low.	STOP and read oil level and add oil.
Top and/or bottom bearing damaged or worn.	Fit new bearings.

7.2.9 Speed too high

Possible cause	Action
Incorrect transmission (50 Hz gear running on 60 Hz power supply).	 DANGER: Disintegration hazard STOP immediately! Install correct transmission. Contact your local Alfa Laval representative. The bowl must be inspected.
Frequency of power supply too high.	Check.

7.2.10 Speed too low

Possible cause	Action
Motor failure.	Repair the motor.
Top/bottom bearings damaged or worn.	Fit new bearings.
Bearing overheated/damaged.	Fit new bearings.
Incorrect transmission (60 Hz gear running on 50 Hz power supply).	Make sure that the gear is intended for 50 Hz power supply.

7.2.11 Water in oil sump

Possible cause	Action
Bowl casing drain obstructed.	Clean. Change oil.
Leakage at top bearing.	Fit a new seal ring and change oil.
Condensation.	Clean the oil sump. Change oil.

7.2.12 Liquid flows through bowl casing drain

Possible cause	Action
Broken water seal.	Stop feed and feed water to create water seal.
Too high throughput	Reduce the feed.
The supply of displacement/sealing water is not sufficient due to clogged strainer, kinked hose or low water pressure.	Straighten the hose or clean the strainer. Make sure the water pressure is 200-600 kPa (2-6 bar).
Seal ring on gravity/clarifier disc defective.	Fit a new seal ring.
Bowl hood seal ring defective.	Fit a new seal ring.
Bowl speed too low.	Make sure current is on. Inspect motor and power transmission.